

Curriculum of Andrea Davini

General Informations

position: Full Professor of Mathematical Analysis, Università di Roma *La Sapienza*,
address: Dipartimento di Matematica “G. Castelnuovo”, Università di Roma *La Sapienza*,
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Positions

- 2025 - today Full Professor, Dipartimento di Matematica “G. Castelnuovo”, Università di Roma *La Sapienza*.
- 2019 - 2025 Associate Professor, Dipartimento di Matematica “G. Castelnuovo”, Università di Roma *La Sapienza* (January 2019 – September 2025).
- 2018 National Scientific Qualification to cover a Full Professor position in Mathematical Analysis (expiring date: July 2029).

Qualification à Professeur des Universités (France) for Sections 25 (Mathématiques) and 26 (Mathématiques appliquées et applications des mathématiques)
- 2007 - 2018 Assistant Professor, Dipartimento di Matematica “G. Castelnuovo”, Università di Roma *La Sapienza*.
- 2006 - 2007 Marie Curie post-doc position (*Marie Curie Intra European Fellowship*) at UMPA, École Normale Supérieure de Lyon (France), (October 2006 – April 2007).
- 2005 - 2006 Post-doc position at the Dipartimento di Matematica Pura ed Applicata, Università di Padova (October 2005 – September 2006).
- 2004 - 2005 Post-doc position at the Dipartimento di Matematica Pura ed Applicata, Università di Pisa (January 2004 – September 2005).

Education

- 2001 - 2003 Ph.D. student in Mathematics at the Dipartimento di Matematica, Università di Pisa (January 2001 – December 2003).
Title of the thesis (defended in May 2004): *Finsler metrics in Optimization Problems and Hamilton-Jacobi equations*.
Advisor: prof. Giuseppe BUTTAZZO
- 1999 - 2000 Fellowship sponsored by Istituto Nazionale di Alta Matematica within a program for doctoral studies abroad. Attending the courses of D.E.A. *Equationes aux Dérivées Partielles et Calcul Scientifique* at the University of Paris 11 (October 1999 – September 2000).

- 1995 - 1999 Undergraduate student in Mathematics at the Università di Pisa.
Degree *cum laude* in Mathematics at the Università di Pisa.
Title of the thesis (defended on September 30, 1999): *Calibrations for minimal surfaces and singular cones*.
Advisor: prof. Giovanni ALBERTI
- 1992 - 1995 Undergraduate student in Engineering, Università di Pisa.

Teaching

- 2022 - 2024 *Istituzioni di Matematica 1* (first-year course in calculus for students of Chemistry, 90 hours) and *Analisi Superiore* (second-year of the Master Degree in Mathematics and Applied Mathematics, 48 hours), Università di Roma *La Sapienza*.
- 2021 - 2022 *Istituzioni di Matematica 1* (first-year course in calculus for students of Chemistry, 120 hours), Università di Roma *La Sapienza*.
- 2020 - 2021 *Istituzioni di Matematica I* (first-year course in calculus for students of Chemistry, 120 hours) and *Istituzioni di Matematica II* (first-year course in calculus for students of Chemistry, 20 hours), *Sapienza* Università di Roma.
- 2019 - 2020 *Istituzioni di Matematica II* (first-year course in calculus for students of Chemistry, 40 hours) and *Analisi Matematica 1* (first-year course of the Bachelor Degree in Mathematics, 90 hours), *Sapienza* Università di Roma.
- 2018 - 2019 *Istituzioni di Matematica II* (first-year course in calculus for students of Chemistry, 60 hours) and *Calcolo integrale* (first-year course of the Bachelor Degree in Computer Science, 60 hours), *Sapienza* Università di Roma.
- 2017 - 2018 *Analisi Reale* (third-year course of Real Analysis for students of Mathematics, 72 hours), Università di Roma *La Sapienza*.
- 2016 - 2017 *Istituzioni di Analisi Superiore* (course of Real Analysis for Master students in Mathematics, 72 hours), Università di Roma *La Sapienza*.
Recitation classes for *Istituzioni di Matematica 1* (first-year course in calculus for students of Chemistry, 36 hours), Università di Roma *La Sapienza*.
- 2015 - 2016 *Istituzioni di Analisi Superiore* (course of Real Analysis for students of Mathematics, 72 hours), Università di Roma *La Sapienza*.
- 2013 - 2015 *Istituzioni di Matematica 2* (second-year course in calculus for students of Architecture, 50 hours), Università di Roma *La Sapienza*.
- 2012 - 2013 *Equazioni Differenziali Nonlineari* (fifth-year course in Nonlinear PDEs for students in Mathematics, 48 hours), Università di Roma *La Sapienza*.
- 2011 - 2012 *Analisi* (first-year course in calculus for students in Physics, 90 hours), Università di Roma *La Sapienza*.
- 2010 - 2011 Recitation classes for *Analisi* (first-year course in calculus for students in Physics), Università di Roma *La Sapienza*.

- 2009 - 2010 *"Istituzioni di Matematica"* (basic calculus for students in Natural Science, 72 hours), Università di Roma *La Sapienza*.
"Omogeneizzazione per equazioni di Hamilton–Jacobi", course for the Ph.D. in Mathematics (30 hours), Università di Roma *La Sapienza*.
- 2008 - 2009 *"Calcolo Differenziale"* (basic calculus for students in Computer Science, 48 hours), Università di Roma *La Sapienza*.
- 2007 - 2008 Recitation classes for *Analisi* (first-year course in calculus for students in Physics), Università di Roma *La Sapienza*.
- 2005 - 2006 Recitation classes for *Matematica 1* (basic calculus for students in Engineering), Università di Padova (course held by Prof. G.P. Leonardi).
- 2004 - 2005 Recitation classes for *Analisi Matematica 1* (basic calculus for students in Engineering), Università di Pisa (course held by Prof. H. Beirão da Veiga).
- 2003 - 2004 Recitation classes for *Analisi Matematica 1* (basic calculus for students in Engineering), Università di Pisa (course held by Prof. H. Beirão da Veiga).

Supervision of students

- T. Thévenet, *Counterexamples to the full convergence for the discounted HJ equations: a detailed account*, internship supervision of a Master's student from École Normale Supérieure–PSL (Paris), February–June 2025.
- L. Biagi, *On the vanishing discount approximation of the Hamilton–Jacobi equation*, Master Thesis in Mathematics, July 2024.
- M. Scappaticci, *Teoria ergodica e ottimizzazione*, Master Thesis in Applied Mathematics, March 2019.
- R. Scarpellino, *Hamilton–Jacobi equations and minimal time function*, Master Thesis in Applied Mathematics, March 2014.
- S. De Angelis, *Metric formulae for Hamilton–Jacobi equations*, Master Thesis in Applied Mathematics, March 2014.
- B. Reatini, *Lebesgue measure theory*, Bachelor Thesis in Mathematics, March 2012.

Research

Mini-courses

- **September 2025** *An Introduction to Viscosity Solutions with Applications in Economics*, (12h), Toulouse School of Economics (France).
- **June 2022.** *Random Lax–Oleinik semigroups for Hamilton–Jacobi systems*, on-line course (8h), (Beijing/Shanghai University) (China).
- **March 2017.** *Weak KAM Theory for Hamilton–Jacobi systems* (4h), Nanjing University (China) (March 9–10, 2017).

- **Gennaio 2017.** *Aubry Theory for systems of weakly coupled Hamilton-Jacobi equations* (4h). In the conference *Beyond Hamilton-Jacobi, Last call to Bordeaux*, Université de Bordeaux (France) (January 9–13, 2017).
- **January 2013.** Aubry sets for weakly coupled systems of Hamilton-Jacobi equations (3h). *Tutorial Workshop on Weak KAM Theory and Related Topics*, The University of Tokyo (Japan) (January 15–18, 2013).
- **October 2010.** *Weak KAM Theory and homogenization of Hamilton-Jacobi equations* (8h), in the thematic semester *Sistemas Dinámicos y geometría: tres aproximaciones*, Instituto de Matemática Interdisciplinar, Universidad Complutense de Madrid, Madrid (Spain) (October 25–29, 2010).

Recent invited talks

2025

- *Stochastic Homogenization of viscous HJ equations in 1d*, Recent progress in Hamilton-Jacobi equations and related topics, June 2-6, 2025, Nanjing
- *Convergence/divergence phenomena in the vanishing discount limit of HJ equations*, Waseda University, 12 April 2025.

2024

- *Stochastic homogenization of viscous HJ equations in 1d*, Università di Padova , 25 October 2024.
- *Stochastic homogenization of viscous HJ equations in 1d*, Seminario P(n), Dipartimento di Matematica, Università di Roma “La Sapienza”, 6 June 2024.
- *Stochastic homogenization of viscous HJ equations in 1d*, 50 anni di Calcolo delle Variazioni, Pisa, 20-22 May 2024.
- *Stochastic homogenization of viscous HJ equations in 1d*, Waseda University, 27 April 2024.

2023

- *Convergence of the solutions of the discounted Hamilton-Jacobi equation*, Continuum mechanics dialogues. On the occasion of Cesare Davini’s 80th birthday, Udine, 4-5 December 2023.
- *On the vanishing discount approximation for compactly supported perturbations of periodic Hamiltonians*, Nonlinear partial differential equations: theory, numerics and applications. In memory of Maurizio Falcone, Roma, 24-26 May 2023.
- *On the vanishing discount approximation for compactly supported perturbations of periodic Hamiltonians*, Dipartimento di Matematica, Università di Roma “La Sapienza”, May 15, 2023.
- *On the vanishing discount approximation for compactly supported perturbations of periodic Hamiltonians*, Recent trends in optimal control and partial differential equations, Pisa, 8-10 May 2023.

2020

- *On the vanishing discount problem from the negative direction*, Nanjing University, (Cina), Zoom seminar, 17 Novembre 2020.

2019

- *The vanishing discount problem for HJ-systems*, New trends in Hamilton-Jacobi: PDE, Control, Dynamical Systems and Geometry, Fudan University, Shanghai (Cina), 1–6 Luglio 2019.

- *Convergenza delle soluzioni per equazioni di Hamilton–Jacobi con fattore di sconto evanescente*, XXI Congresso U.M.I., Pavia, 2–7 settembre 2019.

2018

- *Stochastic homogenization of viscous and non-viscous HJ equations with nonconvex Hamiltonians*, LMS Durham Symposium, Durham (UK), August 20–24, 2018.
- *Weak KAM Theory for Hamilton-Jacobi systems*, Rio ICM Satellite Conference Weak KAM, PUC, Rio de Janeiro (Brazil) , July 23–27, 2018.

2017

- *Homogenization of viscous and non-viscous HJ equations in random media*, Università degli Studi di Padova, May 25, 2017.
- *Convergence of the solutions of discounted Hamilton-Jacobi systems*, Nanjing University, (Cina), March 16, 2017.

2016

- *Convergence of the solutions in the ergodic approximation of the HJ equation*, at the conference *Hamilton-Jacobi Equations: new trends and applications*, Rennes (France), May 2016.
- *Convergence of the solutions of the discounted H-J equation*, Dipartimento di Matematica, Università degli Studi di Roma “Tor Vergata”, May 2016.
- *Convergence of the solutions of the discounted H-J equation*, Dipartimento di Matematica, Università di Roma “La Sapienza”, February 2016.

Invited visits abroad

- Tsuda University, Tokyo (Japan), 21–30 April, 2024. Invited by Prof. Hitoshi ISHII.
- School of Mathematics and Statistics, Beijing Institute of Technology, Beijing (China), 6–20 April, 2024. Invited by Prof. Lin WANG.
- Institut de Mathématiques de Jussieu, Université Pierre et Marie Curie, Paris (France), 26 February – 7 March 2024. Invited by Prof. Maxime ZAVIDOVIQUE.
- CUNY and Baruch College (New York), 18–29 April, 2023. Invited by Prof. Elena KOSYGINA.
- Université Paris Dauphine (France), 11–16 September, 2022. Invited by Prof. Bruno ZILLOTTO.
- Invitation as Research Member to the MSRI Semester *Hamiltonian systems, from topology to applications through analysis* (August 13, 2018 to December 14, 2018, Berkeley) for the period September 13 - December 14, 2018.
- CIMAT at Guanajuato (Mexico), July 4–14, 2017. Invited by Prof. Renato ITURRIAGA.
- Institut de Mathématiques de Jussieu, Université Pierre et Marie Curie, Paris (France), April 3–7, 2017. Invited by dr. Maxime ZAVIDOVIQUE.
- Nanjing University (Cina), March 4–17, 2017. Invited by Prof. Wei CHENG.

- CUNY and Baruch College (New York), March 30–April 6, 2016. Invited by Prof. Elena KOSYGINA.
- Professeur Invité at the Institut de Mathématiques de Jussieu, Université Pierre et Marie Curie, Paris (France), May 5–June 5, 2015.
- CIMAT at Guanajuato (Mexico), March 30–April 28, 2015. Invited by Prof. Gonzalo CONTRERAS.
- Mittag-Leffler Institute, thematic trimester in *Homogenization and Random Phenomenon*, September 17th–November 1st, 2014, Stockholm (Sweden). Invited by Prof. Ari LAPTEV.
- MSI, Australian National University, Canberra (Australia), November 18–December 20, 2013. Invited by Prof. Xu-Jia WANG.
- UMPA, École Normal Supérieure de Lyon, Lyon (France), May 6–10, 2013. Invited by Prof. Albert FATHI.
- Institut de Mathématiques de Jussieu, Université Pierre et Marie Curie, Parigi (France), May 7–18, 2012. Invited by dr. Maxime ZAVIDOVIQUE.
- Unité de mathématiques pures et appliquées, ENS di Lione (France), April 12–16, 2010. Invited by Prof. Albert FATHI.
- IMPA, Rio de Janeiro (Brazil), May 11–30, 2008. Invited by prof. Hermano FRID.

Organization of scientific events

- Organizzatore del Convegno *The Hamilton-Jacobi equation in nonlinear PDEs, dynamics and optimal control: a celebration of Antonio Siconolfi's 70th birthday* (Roma, 5–6 maggio, 2022).
- Co-organizer of the workshop *From Optimal Control to Maximum Principle* (Agropoli, September 12–14, 2018).
- Co-organizer of the INdAM workshop *The Hamilton-Jacobi equation: at the crossroads of PDE, dynamical systems and geometry* (Cortona, June 22–27, 2015).

Publications

- [31] DAVINI, A., SAONA, R., ZILLOTTO, B., Stochastic homogenization of HJ equations: a differential game approach, *Ann. Inst. H. Poincaré Anal. Non Linéaire* (2026), online first.
- [30] DAVINI, A., Stochastic homogenization of quasiconvex degenerate viscous HJ equations in 1d, *Calc. Var. Partial Differential Equations*, 64 (2025), no. 2, Paper No. 38.
- [29] DAVINI, A., GARMENDIA, J. L. P., ITURRIAGA, R., PARDO, J.-L., SÁNCHEZ-MORGADO, H., Discrete approximation of stochastic Mather measures. *Proc. Amer. Math. Soc.* 153 (2025), no. 8, 3319–3333.
- [28] DAVINI, A., KOSYGINA, E., YILMAZ, A., Stochastic homogenization of nonconvex viscous Hamilton-Jacobi equations in one space dimension, *Comm. Partial Differential Equations*, 49 (2024), no. 7–8, 698–734.

- [27] DAVINI, A., Stochastic homogenization of a class of quasiconvex and possibly degenerate viscous HJ equations in 1d, *J. Convex Anal.* 31 (2024), no. 2, 477–496.
- [26] CAPUZZO-DOLCETTA, I., DAVINI, A., On the vanishing discount approximation for compactly supported perturbations of periodic Hamiltonians: the 1d case, *Comm. Partial Differential Equations* 48 (2023), no. 4, 576–622.
- [25] DAVINI, A., KOSYGINA, E., Stochastic homogenization of a class of nonconvex viscous HJ equations in one space dimension, *J. Differential Equations* 333 (2022), 231–267.
- [24] DAVINI, A., ISHII, H., ITURRIAGA, R., SÁNCHEZ-MORGADO, H., Discrete approximation of the viscous Hamilton–Jacobi equation, *Stoch. Partial Differ. Equ. Anal. Comput.* 9 (2021), no. 4, 1081–1104.
- [23] DAVINI, A., WANG, L., On the vanishing discount problem from the negative direction. *Discrete Contin. Dyn. Syst.*, 41 (2021), no. 5.
- [22] DAVINI, A., ZAVIDOVIQUE, M., Convergence of the solutions of discounted Hamilton–Jacobi systems. *Adv. Calc. Var.*, 14 (2021), no. 2, 193–206.
- [21] DAVINI, A., Existence and uniqueness of solutions to parabolic equations with superlinear Hamiltonians. *Commun. Contemp. Math.*, 21 (2019), no. 1.
- [20] DAVINI, A., SICONOLFI, A., ZAVIDOVIQUE, M., Random Lax–Oleinik semigroups for Hamilton–Jacobi systems. *J. Math. Pures Appl.* (9) 120 (2018), 294–333.
- [19] DAVINI, A., KOSYGINA, E., Homogenization of viscous and non-viscous HJ equations: a remark and an application. *Calc. Var. Partial Differential Equations*, 56 (2017), no. 4, 56–95.
- [18] DAVINI, A., FATHI, A., ITURRIAGA, R., ZAVIDOVIQUE, M., Convergence of the solutions of the discounted equation: the discrete case. *Math. Z.* 284 (2016), no. 3–4, 1021–1034.
- [17] DAVINI, A., FATHI, A., ITURRIAGA, R., ZAVIDOVIQUE, M., Convergence of the solutions of the discounted Hamilton–Jacobi equation. *Invent. Math.* 206 (2016), no. 1, 29–55.
- [16] DAVINI, A., SICONOLFI, A. Existence and regularity of strict critical subsolutions in the stationary ergodic setting. *Ann. Inst. H. Poincaré Anal. Non Linéaire* 33 (2016), no. 2, 243–272.
- [15] DAVINI, A., ZAVIDOVIQUE, M., On the (non) existence of viscosity solutions of multi-time Hamilton–Jacobi equations. *J. Differential Equations* 258 (2015), no. 2, 362–378.
- [14] DAVINI, A., ZAVIDOVIQUE, M. Aubry sets for weakly coupled systems of Hamilton–Jacobi equations. *SIAM J. Math. Anal.* 46 (2014), no. 5, 3361–3389.
- [13] DAVINI, A., ZAVIDOVIQUE, M. Weak KAM theory for nonregular commuting hamiltonians. *Discrete Contin. Dyn. Syst. Ser. B* 18 (2013), no. 1, 57–94.
- [12] DAVINI, A., SICONOLFI, A. Weak KAM Theory topics in the stationary ergodic setting. *Calc. Var. Partial Differential Equations* 44 (2012), 3–4, 319–350.
- [11] DAVINI, A., SICONOLFI, A. Metric techniques for convex stationary ergodic Hamiltonians. *Calc. Var. Partial Differential Equations* 40 (2011), , 3–4, 391–421.

- [10] DAVINI, A., SICONOLFI, A. Exact and approximate correctors for stochastic Hamiltonians: the 1-dimensional case. *Math. Ann.* 345 (2009), no. 4, 749–782.
- [9] DAVINI, A. Integral representation of abstract functionals of autonomous type. *Proc. Roy. Soc. Edinburgh Sect. A* 138 (2008), no. 4, 725–754.
- [8] DAVINI, A., PONSIGLIONE, M. Homogenization of two-phase metrics and applications. *J. Anal. Math.* 103 (2007), 157–196.
- [7] DAVINI, A. Bolza problems with discontinuous Lagrangians and Lipschitz-continuity of the value function. *SIAM J. Control Optim.* 46 (2007), no. 5, 1897–1921.
- [6] DAVINI, A., SICONOLFI, A. A generalized dynamical approach to the large time behavior of solutions of Hamilton-Jacobi equations. *SIAM J. Math. Anal.* 38 (2006), no. 2, 478–502.
- [5] BRIANI, A., DAVINI, A. Monge solutions for discontinuous Hamiltonians. *ESAIM Control Optim. Calc. Var.* 11, 2 (2005), 229–251 (electronic).
- [4] DAVINI, A. Smooth approximation of weak Finsler metrics. *Differential Integral Equations* 18 (2005), no. 5, 509–530.
- [3] DAVINI, A. On the relaxation of a class of functionals defined on Riemannian distances. *J. Convex Anal.* 12 (2005), no. 1, 113–130.
- [2] BUTTAZZO, G., DAVINI, A., FRAGALÀ, I., MACIÀ, F. Optimal Riemannian distances preventing mass transfer. *J. Reine Angew. Math.* 575 (2004), 157–171.
- [1] DAVINI, A. On calibrations for Lawson’s cones. *Rend. Sem. Mat. Univ. Padova* 111 (2004), 55–70.

Preprint

- [32] DAVINI, A., ISHII, H., The vanishing discount problem for nonlocal Hamilton-Jacobi equations, *ArXiv e-print* (2025).
- [33] DAVINI, A., Stochastic homogenization of nondegenerate viscous HJ equations in 1d, *ArXiv e-print* (2024).
- [34] DAVINI, A., NI, P., YAN, J., ZAVIDOVIQUE, M., Convergence/divergence phenomena in the vanishing discount limit of Hamilton-Jacobi equations, *ArXiv e-print* (2024).

Last update: April 6, 2026